12th BOARD HACKER

INVERSE TRIGONOMETRIC Functions

INTRODUCTION

$\sin\theta = x$ } → Trigonometry

$\theta = \sin^{-1} x$ } → Inverse Trigonometric Function

Example :-

If $\sin 60^\circ = \frac{\sqrt{3}}{2}$,

Then $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$ or $\frac{\pi}{3}$

Similarly, $\cos^{-1}\frac{\sqrt{3}}{2} = 30^\circ$ or $\frac{\pi}{6}$

Important

↳ $\sin\theta = \frac{1}{2}$ for $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$ .... etc.

But $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$ only

Because

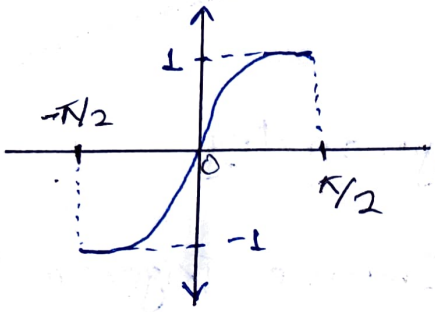
Inverse has a unique value

Inverse Trig Functions give values in the principal domain only.

GRAPHS OF ITF

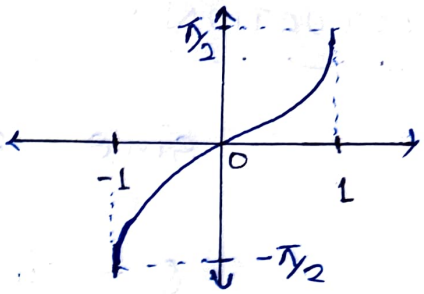
1) INVERSE SINE FUNCTION

$y = \sin x$



Inverse

$y = \sin^{-1}(x)$



Domain → $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

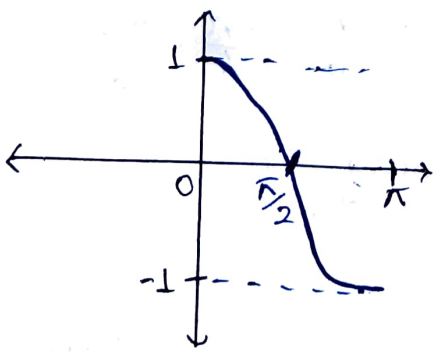
Range → $[-1, 1]$

Domain → $[-1, 1]$

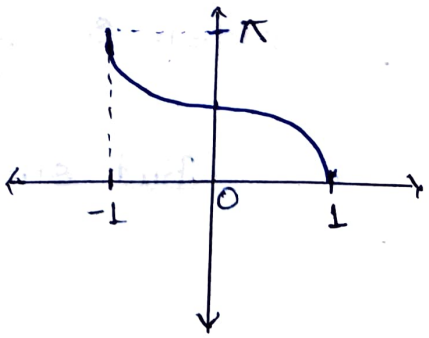
Range → $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2) INVERSE COSINE FUNCTION

$y = \cos x$



$y = \cos^{-1}(x)$



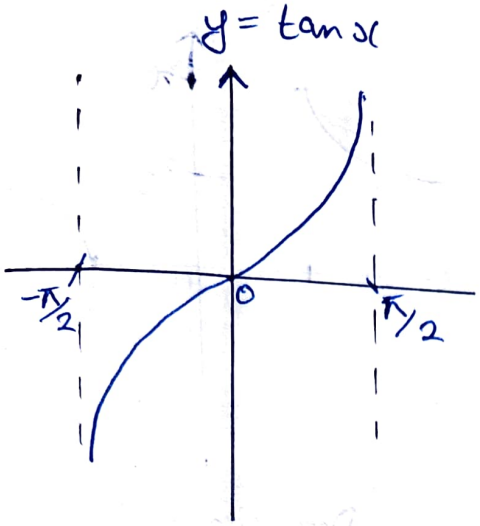
Domain → $[0, \pi]$

Range → $[-1, 1]$

Domain → $[-1, 1]$

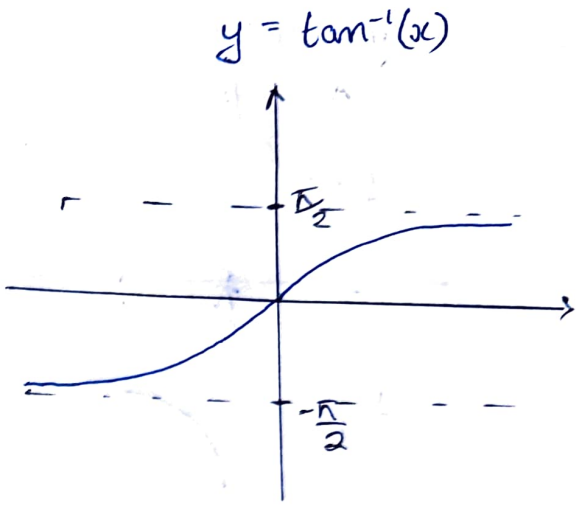
Range → $[0, \pi]$

3) INVERSE TANGENT FUNCTION



Domain → $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

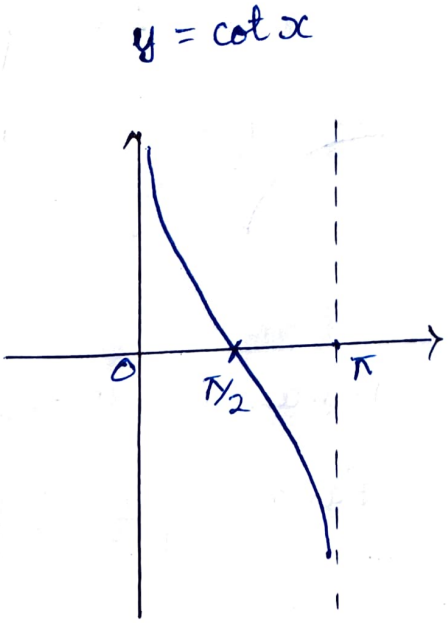
Range → $(-\infty, \infty)$



Domain → R or $(-\infty, \infty)$

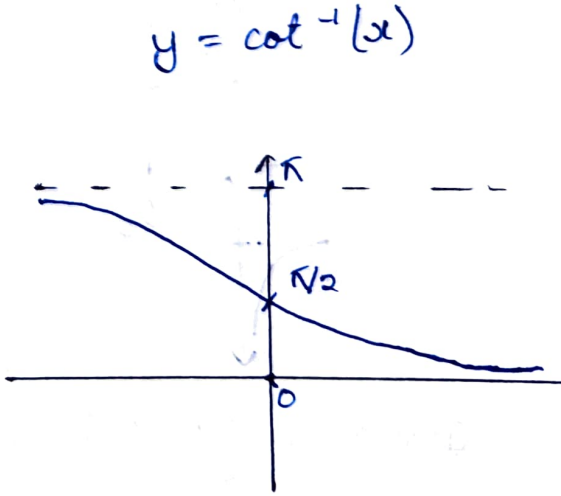
Range → $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$

4) INVERSE COTANGENT FUNCTION



Domain → $(0, \pi)$

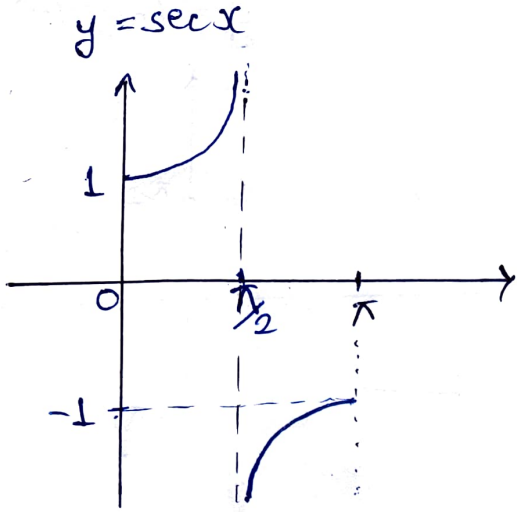
Range → $(-\infty, \infty)$



Domain → $(-\infty, \infty)$

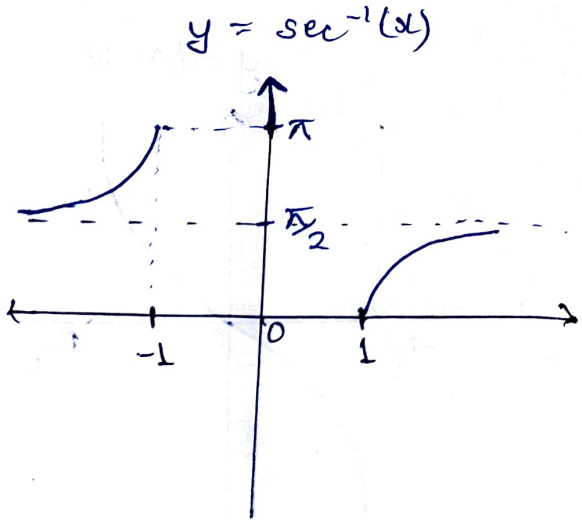
Range → $(0, \pi)$

5) INVERSE SECANT FUNCTION



Domain → $[0,\pi] - \{\frac{\pi}{2}\}$

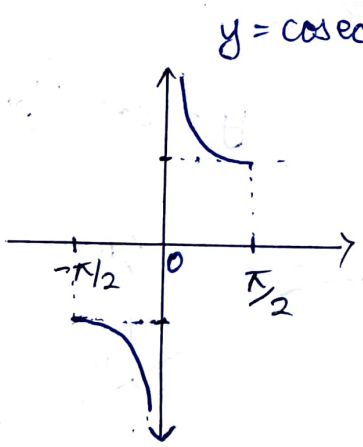
Range → $(-\infty,-1] \cup [1,\infty)$



Domain → $(-\infty,-1] \cup [1,\infty)$

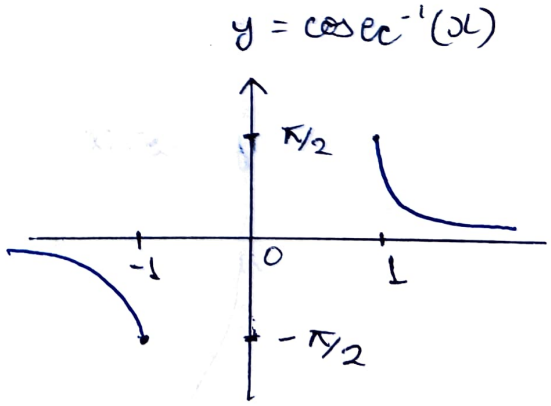
Range → $[0,\pi] - \{\frac{\pi}{2}\}$

6) INVERSE COSECANT FUNCTION



Domain $\in \left[-\frac{\pi}{2},\frac{\pi}{2}\right] - \{0\}$

Range $\in (-\infty,-1] \cup [1,\infty)$



Domain $\in (-\infty,-1] \cup [1,\infty)$

Range $\in \left[-\frac{\pi}{2},\frac{\pi}{2}\right] - \{0\}$

PROPERTIES OF ITF

	Domain
1. $\sin^{-1}(-x) = -\sin^{-1}(x)$	$x \in [-1, 1]$
2. $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}(x)$	$x \in (-\infty, -1] \cup [1, \infty)$
3. $\tan^{-1}(-x) = -\tan^{-1}(x)$	$x \in (-\infty, \infty)$
4. $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$	$x \in -1 \leq x \leq 1$
5. $\sec^{-1}(-x) = \pi - \sec^{-1}(x)$	$x \in (-\infty, -1] \cup [1, \infty)$
6. $\cot^{-1}(-x) = \pi - \cot^{-1}(x)$	$x \in (-\infty, \infty)$

Example :- What is the value of $\cos^{-1}\left(-\frac{1}{2}\right)$?

$$\Rightarrow \cos^{-1}(-x) = \pi - \cos^{-1}(x)$$

$$\Rightarrow \cos^{-1}\left(-\frac{1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right)$$

$$= \pi - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

RECIPROCAL PROPERTIES

1. $\sin^{-1}\frac{1}{x} = \operatorname{cosec}^{-1} x$
2. $\operatorname{cosec}^{-1}\frac{1}{x} = \sin^{-1} x$
3. $\cos^{-1}\frac{1}{x} = \sec^{-1} x$
4. $\sec^{-1}\frac{1}{x} = \cos^{-1} x$
5. $\tan^{-1}\frac{1}{x} = \cot^{-1} x$
6. $\cot^{-1}\frac{1}{x} = \tan^{-1} x$ for $x > 0$
7. $\cot^{-1} x = \pi + \tan^{-1}\frac{1}{x}$ for $x < 0$

COMPLEMENTARY PROPERTIES

1. $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ $\quad x \in [-1, 1]$

2. $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$ $\quad x \in R$

3. $\text{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}$ $\quad x \in (-\infty, -1] \cup [1, \infty)$

Example :- If $4\sin^{-1}x + \cos^{-1}x = \pi$ then find x.

Ans $\Rightarrow 4\sin^{-1}x + \cos^{-1}x = \pi$

$\Rightarrow 3\sin^{-1}x + \underbrace{\sin^{-1}x + \cos^{-1}x}_{\pi/2} = \pi$

$\Rightarrow 3\sin^{-1}x + \frac{\pi}{2} = \pi$

$\Rightarrow 3\sin^{-1}x = \frac{\pi}{2}$

$\Rightarrow \sin^{-1}x = \frac{\pi}{6}$

$\Rightarrow x = \sin\frac{\pi}{6}$

$= \frac{1}{2}$ Ans

CONVERSION OF ITF

	Domain	Range
1) $\sin(\sin^{-1}x) = x$	$x \in [-1, 1]$	$x \in [-\pi/2, \pi/2]$
2) $\cos(\cos^{-1}x) = x$	$x \in [-1, 1]$	$x \in [0, \pi]$
3) $\tan(\tan^{-1}x) = x$	$x \in (-\infty, \infty)$	$x \in (-\frac{\pi}{2}, \frac{\pi}{2})$
4) $\cot(\cot^{-1}x) = x$	$x \in (-\infty, \infty)$	$x \in (0, \pi)$
5) $\sec(\sec^{-1}x) = x$	$x \in (-\infty, -1] \cup [1, \infty)$	$x \in [0, \pi] - \{\frac{\pi}{2}\}$
6) $\operatorname{cosec}(\operatorname{cosec}^{-1}x) = x$	$x \in (-\infty, 1] \cup [1, \infty)$	$x \in [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Q Find the value of $\sin(\tan^{-1}x)$.

Ans $\Rightarrow$ $\tan^{-1}\left(\frac{x}{1}\right)$ $\quad \frac{P}{B} = \frac{x}{1} \longrightarrow H = \sqrt{x^2+1}$ ... i)

Let $\tan\alpha = \frac{x}{1}$

$\therefore \sin\alpha = \frac{x}{\sqrt{x^2+1}}$ (from eq. i))

$\therefore \alpha = \tan^{-1}x = \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)$

So, $\sin(\tan^{-1}x) = \sin\left(\sin^{-1}\frac{x}{\sqrt{x^2+1}}\right)$

$= \frac{x}{\sqrt{x^2+1}}$ (from conversion property)

Ans

SHORTCUT OF ITF CONVERSION

$$\sin^{-1}\left(\frac{P}{H}\right) = \cos^{-1}\left(\frac{B}{H}\right) = \tan^{-1}\left(\frac{P}{B}\right)$$
$$= \sec^{-1}\left(\frac{H}{B}\right) = \cot^{-1}\left(\frac{B}{P}\right)$$
$$= \operatorname{cosec}^{-1}\left(\frac{H}{P}\right)$$

Ex: $\cos\left(\tan^{-1}\left(\frac{5}{12}\right)\right)$

$P = 5$
$B = 12$
$\therefore H = \sqrt{25+144} = 13$

$$\therefore \tan^{-1}\frac{P}{B} = \cos^{-1}\frac{B}{H} = \cos^{-1}\frac{12}{13}$$

So $\tan^{-1}\frac{5}{12} = \cos^{-1}\frac{12}{13}$

$$\therefore \cos\left(\tan^{-1}\frac{5}{12}\right) = \cos\left(\cos^{-1}\frac{12}{13}\right)$$
$$= \frac{12}{13} \text{ Ans}$$

Remember that all the answers of ITF lie in the principal range only

Ex: $\sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

SOME TANGENT PROPERTIES

1) $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ $\quad xy < 1$

$= \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ $\quad xy > 1$

$= \frac{\pi}{2}$ $\quad \because xy = 1$

2) $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\frac{x-y}{1+xy}$, **xy > -1**

3) $\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3) = \pi$

4) $\tan^{-1}(1) + \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{2}$

5) $2\tan^{-1}(x) = \sin^{-1}\frac{2x}{1+x^2}$, $|x| \leq 1$

$= \cos^{-1}\frac{1-x^2}{1+x^2}$, $x \geq 0$

$= \tan^{-1}\frac{2x}{1-x^2}$, $-1 < x < 1$

Q Write the following in simplest form :-

$$\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right), \quad x \neq 0$$

Sol : Let $x = \tan\theta \Rightarrow \theta = \tan^{-1} x$

$$\Rightarrow \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{1}{\cos\theta} - 1}{\frac{\sin\theta}{\cos\theta}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{2\sin^2\theta/2}{2\sin\theta/2\cos\theta/2}\right)$$

$$\Rightarrow \tan^{-1}\left[\tan\frac{\theta}{2}\right]$$

$$\Rightarrow \frac{\theta}{2} = \frac{\tan^{-1}x}{2}$$ Ans

Q Prove that $\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13} = \cos^{-1}\frac{33}{65}$

Sol $\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13}$

$= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{5}{12}$

$= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{5}{12}}{1-\frac{3}{4}\times\frac{5}{12}}\right)$

$= \tan^{-1}\frac{56}{33}$ → P, → B

$\therefore H = \sqrt{56^2+33^2}$

$= 65$

$= \cos^{-1}\frac{33}{65}$

Hence Proved

Q The smallest and largest values of $\tan^{-1}\left(\frac{1-x}{1+x}\right)$, $0 \le x \le 1$ are :-

Sol Let $x = \tan\theta$, $\theta = \tan^{-1}x$

$\Rightarrow \tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right) = \tan^{-1}\left[\tan\left(\frac{\pi}{4}-\theta\right)\right]$

$= \frac{\pi}{4} - \theta = \frac{\pi}{4} - \tan^{-1}x$

As given, $0 \le x \le 1$, So, $\tan^{-1}0 \le \tan^{-1}x \le \tan^{-1}1$

$\Rightarrow 0 \le \tan^{-1}x \le \frac{\pi}{4}$

$\Rightarrow 0 \ge -\tan^{-1}x \ge -\frac{\pi}{4}$

$+\frac{\pi}{4}$ $+\frac{\pi}{4}$ $+\frac{\pi}{4}$

$\therefore \frac{\pi}{4} - \tan^{-1}x \in \left[0, \frac{\pi}{4}\right]$ // Ans

IMPORTANT TIP

IN ORDER TO MAKE INVERSE TRIGONOMETRIC FUNCTION STRONG, WE MUST HAVE A STRONG COMMAND OVER ALL THE FORMULAS OF TRIGONOMETRY (CLASS XI)

Trigonometry Formulas

Function Relationships

$$\sin\theta = \frac{1}{\csc\theta} \qquad \csc\theta = \frac{1}{\sin\theta}$$
$$\cos\theta = \frac{1}{\sec\theta} \qquad \sec\theta = \frac{1}{\cos\theta}$$
$$\tan\theta = \frac{1}{\cot\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$
$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

Opposite Angle Formulas

$$\sin(-\theta) = -\sin(\theta)$$
$$\cos(-\theta) = \cos(\theta)$$
$$\tan(-\theta) = -\tan(\theta)$$
$$\cot(-\theta) = -\cot(\theta)$$
$$\sec(-\theta) = \sec(\theta)$$
$$\csc(-\theta) = -\csc(\theta)$$

Cofunction Formulas (in Quadrant I)

$$\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right) \qquad \cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$$
$$\tan\theta = \cot\left(\frac{\pi}{2} - \theta\right) \qquad \cot\theta = \tan\left(\frac{\pi}{2} - \theta\right)$$
$$\sec\theta = \csc\left(\frac{\pi}{2} - \theta\right) \qquad \csc\theta = \sec\left(\frac{\pi}{2} - \theta\right)$$

Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1$$
$$\tan^2\theta + 1 = \sec^2\theta$$
$$\cot^2\theta + 1 = \csc^2\theta$$

Half Angle Formulas

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}$$
$$\cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}$$
$$\tan\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \frac{1-\cos\theta}{\sin\theta} = \frac{\sin\theta}{1+\cos\theta}$$

Angle Addition Formulas

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$
$$\sin(A-B) = \sin A\cos B - \cos A\sin B$$
$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$
$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$
$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$
$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A\tan B}$$

Double Angle Formulas

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$$
$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

Product-to-Sum Formulas

$$\sin A\cdot\sin B = \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]$$
$$\cos A\cdot\cos B = \frac{1}{2}\left[\cos(A-B) + \cos(A+B)\right]$$
$$\sin A\cdot\cos B = \frac{1}{2}\left[\sin(A+B) + \sin(A-B)\right]$$
$$\cos A\cdot\sin B = \frac{1}{2}\left[\sin(A+B) - \sin(A-B)\right]$$

Triple Angle Formulas

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$
$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

Power Reducing Formulas

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$
$$\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$$

Sum-to-Product Formulas

$$\sin A + \sin B = 2\cdot\sin\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)$$
$$\sin A - \sin B = 2\cdot\sin\left(\frac{A-B}{2}\right)\cdot\cos\left(\frac{A+B}{2}\right)$$
$$\cos A + \cos B = 2\cdot\cos\left(\frac{A+B}{2}\right)\cdot\cos\left(\frac{A-B}{2}\right)$$
$$\cos A - \cos B = -2\cdot\sin\left(\frac{A+B}{2}\right)\cdot\sin\left(\frac{A-B}{2}\right)$$

Arc Length

$$S = r\theta$$

Law of Sines

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left[\frac{1}{2}(A-B)\right]}{\tan\left[\frac{1}{2}(A+B)\right]}$$

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Mollweide's Formulas

$$\frac{a+b}{c} = \frac{\cos\left[\frac{1}{2}(A-B)\right]}{\sin\left(\frac{1}{2}C\right)}$$
$$\frac{a-b}{c} = \frac{\sin\left[\frac{1}{2}(A-B\right]}{\cos\left(\frac{1}{2}C\right)}$$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta = \operatorname{cis}\theta$$

DeMoivre's Formula

$$(r\operatorname{cis}\theta)^n = r^n\operatorname{cis}(n\theta)$$

Polar Multiplication and Division

Let: $a = r_1\operatorname{cis}\theta$ $\qquad b = r_2\operatorname{cis}\varphi$

$$a\cdot b = r_1r_2\operatorname{cis}(\theta+\varphi) \qquad \frac{a}{b} = \frac{r_1}{r_2}\operatorname{cis}(\theta-\varphi)$$